higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING SYSTEMS
NQF LEVEL 2

(6021022)

18 November 2019 (Y-Paper)
13:00–16:00

This question paper consists of 9 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only BLUE or BLACK ink.
6. Write neatly and legibly.

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QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

1.1.1 Device used in V-belt drives to maintain the correct tension and arc of contact:

A Drive pulley
B Idler pulley
C Driven pulley
D Idler sprocket

1.1.2 Yellow and green wire in a three-pin electrical plug:

A Live
B Neutral
C Earth
D Resistor

1.1.3 Component used to trap impurities, such as rust, dirt and dust, in a pneumatic system:

A Filter
B Flexible hose
C Reservoir
D Cylinder

1.1.4 Act of oiling machines with a slippery fluid to reduce friction between parts so that they last longer:

A Vibration
B Alignment
C Lubrication
D Deflection

1.1.5 Component that controls the voltage and current in an electrical circuit:

A Resistor
B Potential difference
C Force
D Work done

1.1.6 Valve used to protect components in a hydraulic system against excessive oil pressure:

A Spool valve
B Rotary valve
C Control valve
D Pressure relief valve

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1.1.7 Free movement or play between a screw thread and nut:

A Displacement ratio
B Effort
C Mechanical advantage
D Backlash

1.1.8 Document that details the work done on a machine and costs involved:

A Manufacturer's manual
B Job card
C Service manual
D Health and safety manual

1.1.9 Stopping of machine and preventing anyone from accidentally switching it on:

A Service
B Isolate
C Tolerance
D Misalignment

1.1.10 Used to reduce friction and remove heat caused by friction during machining of metals on lathe, drilling and milling machines:

A Cutting fluids
B Bolts
C Gaskets
D Locking plates

(10 × 1) (10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 A lever is a simple example of mechanical advantage.

1.2.2 The unit in which to measure pressure is Newton.

1.2.3 PNP transistors convert alternating current to direct current.

1.2.4 Stepper motors are used in the paper feed of computer printers.

1.2.5 Downtime in the workplace leads to the loss of production.

(5 × 1) (5)

[15]

QUESTION 2

2.1 The International System of Units (SI) has basic units.

Give the SI units for each of the following:

2.1.1 Speed

2.1.2 Mass

2.1.3 Power

2.1.4 Area

2.1.5 Electric current

(5 × 1) (5)

2.2 Draw the symbols for each of the following hydraulic components:

2.2.1 Line connection

2.2.2 Check valve

2.2.3 Hydraulic motor

2.2.4 Open tank

2.2.5 Double-acting cylinder

(5 × 1) (5)

2.3 List THREE consequences of wiring an electrical plug incorrectly. (3)

2.4 The purpose of circuit protective devices is to prevent fires and damage to the circuit, and to protect against personal injury.

List FOUR protection devices used in electrical circuits. (4)

2.5 FIGURE 1 shows THREE resistors of 5 Ω, 6 Ω and 7 Ω connected in parallel across a 12 V battery.

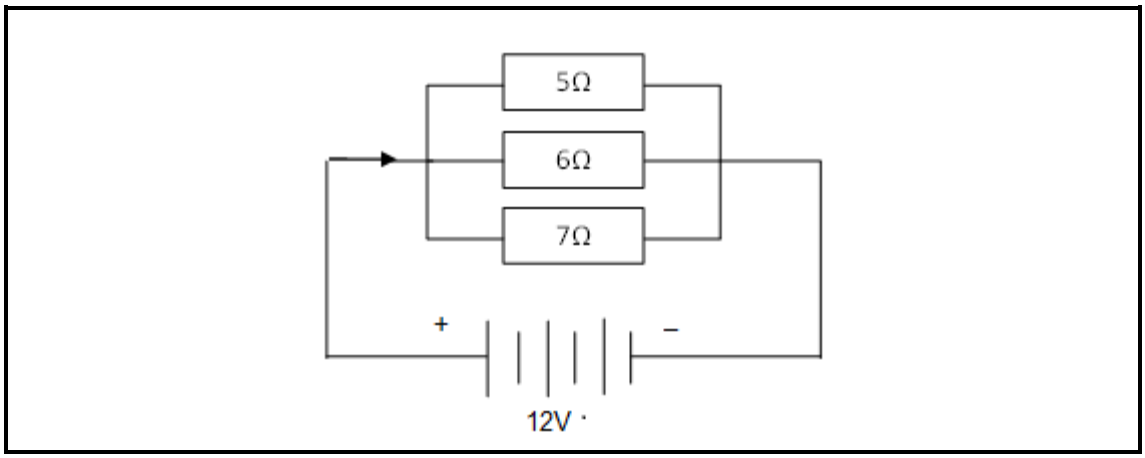


FIGURE 1

2.5.1 Calculate the total resistance of the circuit. (3)

2.5.2 Calculate the total current through the circuit. (2)

2.5.3 Calculate the power used by the 6 Ω resistor.

Hint: $[\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}]$; $[V = I \times R]$; $[P = V \times I]$ (3)

[25]

QUESTION 3

3.1 FIGURE 2 shows a V-belt drive system. The shaft of the motor is connected to Pulley A.

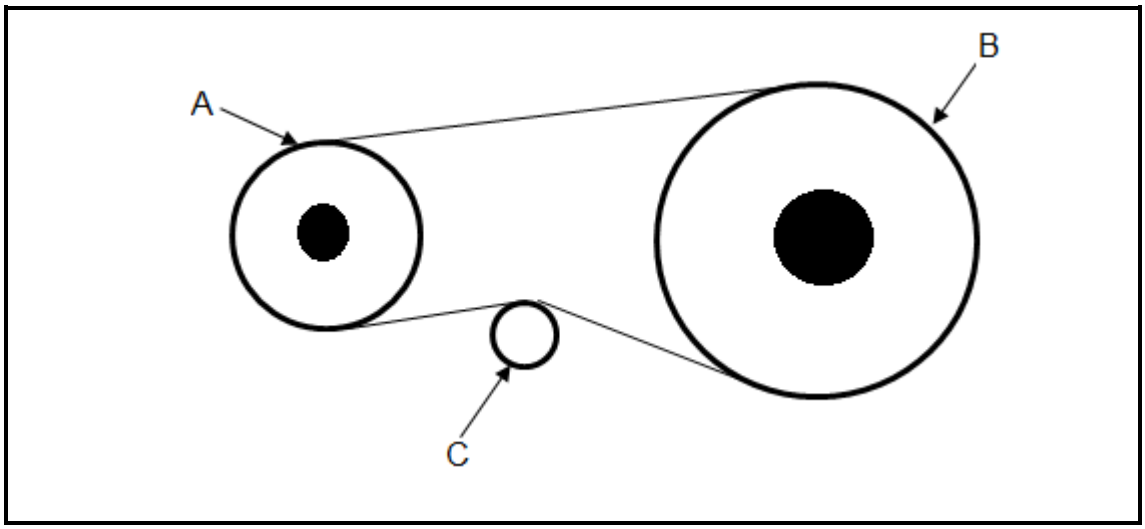


FIGURE 2

3.1.1 Label pulleys A, B and C. (3)

3.1.2 State ONE function of component C. (1)

3.1.3 A motor is connected to Pulley A and it rotates at 1 800 r/min with a 250 mm diameter. Pulley B rotates at 750 r/min.

Calculate the diameter of Pulley B. [Hint: $d \times .n = D \times N$] (3)

3.2 FIGURE 3 shows a simple gear system. An input motor of power 12 kW rotates at 630 r/min and is connected to Gear P with 30 teeth. The gear ratio is 3, 5:1.

Assume that no losses occur.

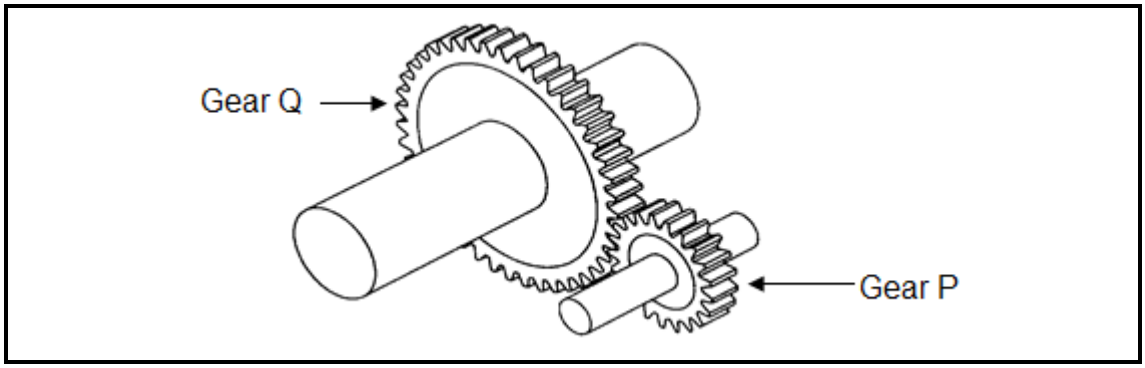


FIGURE 3

3.2.1 Calculate the number of teeth of Gear Q. (2)

3.2.2 Calculate the rotational speed of Gear Q. (2)

3.2.3 The torque developed at Gear P. [Hint: $P = \frac{2\pi NT}{60}$] (3)

3.3 Give ONE pre-check that must be carried out on each of the following hydraulic components:

3.3.1 Reservoirs

3.3.2 Cylinders

3.3.3 Pipes

(3 × 1) (3)

3.4 Chain drives are designed with covers fitted around the moving components.

Explain why chain drive covers are important. (3)

[20]

QUESTION 4

4.1 A control system is an essential part of the operation of any machine or equipment. Computer-controlled machines use a microprocessor to program complicated steps.

Give the meaning of each of the following abbreviations:

4.1.1 CNC

4.1.2 PLC

4.1.3 IC

4.1.4 CPU

(4 × 1) (4)

4.2 Input is the information supplied to a computer by a sensor.

Name FIVE types of sensors used in control systems. (5)

4.3 Machines are easily and reliably controlled by using a variety of switches and micro switches.

State TWO advantages of a micro switch. (2)

4.4 Name FOUR types of braking systems. (4)

4.5 Give FIVE implications of non-conformance with the manufacturer's specifications when setting equipment with simple control systems. (5)

[20]

QUESTION 5

5.1 Name the tool/equipment that is used to perform each of the following tasks:

5.1.1 Measure the outside diameter of a shaft.

5.1.2 Install a bearing on a shaft.

5.1.3 Tighten bolts and nuts to a specified tightness.

5.1.4 Remove a bearing from a shaft.

(4 × 1) (4)

5.2 Maximum chain drive performance requires proper maintenance.

List FIVE maintenance checks to be done on chain drives. (5)

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5.3 Lubrication is critical for the reduction of friction wear between two moving surfaces.

List FOUR oil lubrication accessories. (4)

5.4 Give THREE reasons why it is important to record on the job card the task performed on equipment during maintenance. (3)

5.5 An organisation tries to avoid failure by maintaining its physical resources, such as machinery and equipment.

Explain the difference between *preventive* and *routine maintenance*. (4)
[20]

TOTAL: 100

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